Evaluating Climate Skeptic Arguments

Group: _____ Names: _____

Each group receives five common skeptic arguments that share a structure. Answer questions (a) to (d) for your group only.

Group A — Magnitude: “The numbers are too small to matter”

Your arguments:

- CO_2 is just a trace gas
- Human CO_2 emissions are small compared to natural emissions
- Volcanoes emit more CO_2 than humans
- CO_2 is just a small part of the greenhouse effect
- China and India emit more, so our emissions do not matter

(a) Which of the five is the strongest? One sentence on why.

(b) Which is the weakest? One sentence on why.

(c) Each of these arguments compares a quantity against the wrong reference. For each argument, write two short lines: what it compares, and what the appropriate comparison would be.

(d) Taking the argument you identified as strongest: state the number that settles it, name its source, and identify the one assumption within that number which a skeptic could still legitimately question.

Group B — Causation: “The cause runs the other way”

Your arguments:

- CO_2 lags temperature
- Water vapour is the most powerful greenhouse gas
- The ocean is releasing CO_2
- It is the sun
- It is cosmic rays

(a) Which of the five is the strongest? One sentence on why.

(b) Which is the weakest? One sentence on why.

(c) Each argument either reverses the direction of causation or mistakes a feedback for a cause. For each, state which way the relationship actually runs, and name the evidence that establishes the direction.

(d) Taking the argument you identified as strongest: state the number that settles it, name its source, and identify the one assumption within that number which a skeptic could still legitimately question.

Group C — Selective intervals

Your arguments:

- It has not warmed since 1998
- It is cooling
- 1934 was the hottest year on record
- The Medieval Warm Period was warmer than today
- We are simply recovering from the Little Ice Age

(a) Which of the five is the strongest? One sentence on why.

(b) Which is the weakest? One sentence on why.

(c) Each argument depends on a choice of start point, end point, or region. For each, name the choice being made and explain why that particular choice was selected rather than another.

(d) Taking the argument you identified as strongest: state the number that settles it, name its source, and identify the one assumption within that number which a skeptic could still legitimately question.

Group D — Physical limits

Your arguments:

- The CO_2 effect is saturated
- Climate sensitivity is low
- Clouds provide negative feedback
- The greenhouse effect violates the second law of thermodynamics
- Models are unreliable

(a) Which of the five is the strongest? One sentence on why.

(b) Which is the weakest? One sentence on why.

(c) Four of these claims are true within some regime and fail outside it; one is wrong in every regime. For each, state the conditions under which the claim holds and the conditions under which it fails. Identify the one that never holds.

(d) Taking the argument you identified as strongest: state the number, figure, or physical principle that settles it, name its source, and identify the one assumption a skeptic could still legitimately question.

Group E — Consequence: “It does not matter anyway”

Your arguments:

- CO_2 is plant food
- It is not bad
- Animals and plants can adapt
- It is too expensive to fix
- Pakistan emits less than 1% of global CO_2

(a) Which of the five is the strongest? One sentence on why.

(b) Which is the weakest? One sentence on why.

(c) These arguments concern costs and values as much as physical science. For each, name who bears the cost under the skeptic’s framing, and who bears it in practice.

(d) Taking the argument you identified as strongest: state the evidence that settles it and name its source. Then identify the point at which the argument ceases to be a factual claim and becomes a claim about obligation, that is, about what is owed, and by whom.